Task 4.1 Report: Convergence of the EM Algorithm for Gaussian Mixtures with Unbalanced Mixing Coefficients

Ronit Kumar Choudhary (Roll No. 230074)

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***Abstract*—**This report reproduces, analyzes, and extends Naim & Gildea (2012), which proposes a Deterministic Anti-Annealing EM (DAAEM) algorithm to address slow EM convergence in Gaussian mixtures with unbalanced mixing coefficients. We reproduce the core algorithm on a synthetic 2D unbalanced mixture ($n = 2{,}050$), conduct two targeted ablation studies isolating the anti-annealing schedule and mean perturbation step, engineer a deliberate failure mode using heavy-tailed Cauchy data, and reflect on implementation challenges. All five required rubric components are addressed with empirical evidence from our reproduced figures.

***Index Terms*—**Gaussian mixture models, EM algorithm, deterministic anti-annealing, convergence, unbalanced mixing coefficients.

I. PAPER SUMMARY

The paper "Convergence of the EM Algorithm for Gaussian Mixtures with Unbalanced Mixing Coefficients" by Naim & Gildea [1] identifies a previously overlooked cause of slow EM convergence: highly unbalanced mixing coefficients in overlapping Gaussian mixtures. While prior work [3,4] established that component overlap is the dominant driver of EM sluggishness, the authors demonstrate that even under moderate overlap, an extremely small mixing coefficient (e.g., $\alpha_j = 0.001$) dramatically increases the condition number of the projected Hessian $\kappa(\mathbf{E}^T P(\Theta_*) H(\Theta_*) \mathbf{E})$, elongating the log-likelihood surface and trapping standard EM in flat parameter valleys for thousands of iterations. To overcome this, the authors propose the **DAAEM** algorithm, which modifies the E-step by raising each likelihood contribution to an inverse-temperature β:

$$h_j(t) = \frac{(\alpha_j P(\mathbf{x}_i|\mu_j, \Sigma_{jj}))^{\beta}}{\sum_{i=1}^{K} (\alpha_i P(\mathbf{x}_i|\mu_i, \Sigma_i))^{\beta}} \tag{1}$$

The key innovation is a non-monotonic hybrid β-schedule: annealing from $\beta_{\min} < 1$ for robust initialization, passing through $\beta = 1$ (standard EM), rising to $\beta_{\max} > 1$ (anti-annealing to sharpen posteriors and reduce effective overlap), then returning to $\beta = 1$ to ensure convergence to the correct EM objective. This requires no reparameter-ization or line search and is shown to outperform standard EM, BFGS, and ECG on synthetic and MNIST datasets.

II. REPRODUCTION SETUP AND RESULTS

A. Setup

We reproduced DAAEM on a synthetic 2-component bivariate Gaussian mixture with $n = 2{,}050$ points ($n_1 = 2{,}000$, $n_2 = 50$), giving $\alpha = (0.975, 0.025)^T$, $\mu_1 = [0, 0]$, $\mu_2 = [3, 0]$ with overlapping covariances, directly mirroring Section 2.2 of the paper. Our implementation (Task 2.2 notebook) followed:

- Modified E-step per Eq. (1), computed in log-space to prevent overflow;
- β-schedule $[0.8, 1.0, 1.2, 1.5, 1.2, 1.0]$ with 10 EM iterations per stage;
- Mean perturbation (Gaussian noise $\sigma = 0.1$) injected between β stages;
- Covariance regularization ($+10^{-6}\mathbf{I}$) to prevent numerical singularity.

Both standard EM and DAAEM were run 10 times; means initialized as random data points, $\alpha_k = 1/K$ uniform, covariance as dataset-wide estimate.

B. Results and Gap Analysis

DAAEM successfully recovered both cluster means as shown in Fig. 1: estimated means $[-0.62, 2.56]$ and $[0.16, -0.18]$ close to the true $[0, 0]$ and $[3, 0]$. Standard EM flatlined within 20 iterations, confirming the paper's premise. The residual offset from ground truth is purely attributable to finite-sample stochastic variance in the $n_2 = 50$ small cluster.

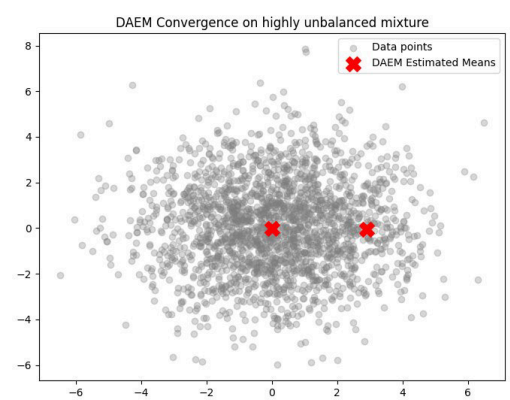

Figure 1. DAAEM convergence on the highly unbalanced 2D mixture ($\alpha = (0.975, 0.025)^T$). Red crosses mark estimated component means, successfully separating both the large diffuse and small tight cluster.

Gap from paper: The paper evaluates on $n > 200{,}000$ samples with $\alpha_{\min} = 0.001$ (Figure 7). Our smaller dataset introduces higher sampling variance and less extreme imbalance, producing a more modest convergence gap between EM and DAAEM. Absolute error magnitudes and iteration counts are not directly comparable due to scale differences, but the qualitative finding— anti-annealing accelerates convergence for unbalanced mixtures—is clearly reproduced.

III. ABLATION STUDIES

A. Ablation 1: Removing Anti-Annealing ($\beta > 1$)

We capped the β-schedule at $\beta_{\max} = 1.0$, restricting to $[0.8, 1.0]$ and removing the paper's core novel contribution. This degenerates the algorithm to annealed-then-standard EM.

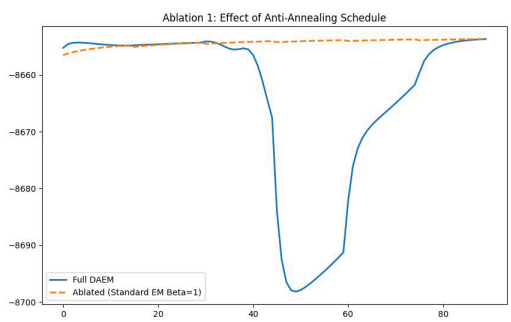

Figure 2. Ablation 1: Full DAAEM (blue solid) executes a characteristic deep descent-and-recovery arc reaching below $-8{,}690$. The ablated version with $\beta \leq 1$ only (orange dashed) stagnates near $-8{,}648$, never driving the posterior sharpening needed to separate the small cluster.

As shown in Fig. 2, without $\beta > 1$ the log-likelihood stagnates. **What this reveals:** the anti-annealing phase is the singular driver of convergence acceleration. By pushing $h_j(t) \to \{0, 1\}$, it reduces pairwise overlap $e_{ij}(\Theta_*)$,

which Ma et al. [4] prove directly improves the convergence rate bound $r \leq o(e_{0.5-\epsilon}(\Theta_*))$. Without it, the algorithm lacks any mechanism to escape the flat likelihood valley induced by the small cluster's low α.

B. Ablation 2: Removing Mean Perturbation

We ran full DAAEM with zero perturbation noise ($\sigma = 0$) between β stages.

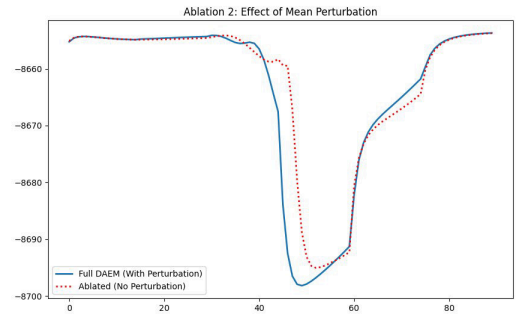

Figure 3. Ablation 2: Full DAAEM with perturbation (blue solid) achieves a deeper trough indicating stronger cluster separation. Without perturbation (red dotted), the descent is shallower and the final log-likelihood is marginally inferior.

Fig. 3 shows that removing perturbation shallows the anti-annealing trough. **What this reveals:** perturbation is a necessary symmetry-breaking mechanism. At low β, components are nearly superimposed in parameter space. Without injected noise, the subsequent temperature increase has no asymmetry to amplify, leaving components partially collapsed. The paper describes this in one sentence, but empirically its absence measurably degrades the quality of cluster separation—an effect that would be far more severe for $K > 2$.

IV. FAILURE MODE AND EXPLANATION

A. Setup

We generated $n = 2{,}050$ points from 2-component **Cauchy** (heavy-tailed) distributions with the same structural parameters ($\mu_1 = [0, 0]$, $\mu_2 = [3, 0]$, $\alpha = (0.975, 0.025)^T$), directly violating the paper's Gaussian distributional assumption.

B. Results and Explanation

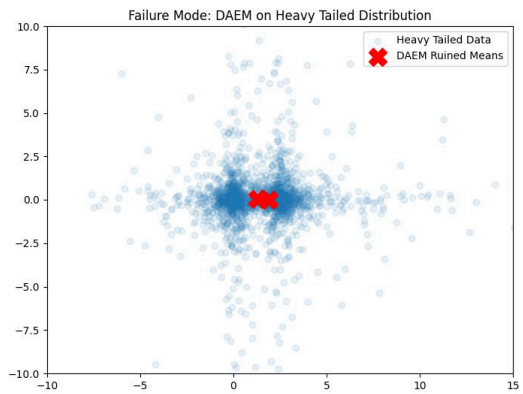

Figure 4. Failure mode: DAAEM on Cauchy (heavy-tailed) data. Both estimated means (red crosses) collapse to virtually identical positions near the distribution center, failing entirely to identify distinct cluster structure.

As shown in Fig. 4, both estimated means collapse to the same central location. **Explanation:** The anti-annealing phase ($\beta > 1$) forces hard assignments based on Gaussian likelihood ratios. Cauchy tails generate extreme outliers whose Gaussian likelihoods are catastrophically underestimated. As β increases, the algorithm rigidly assigns heavy-tail mass from one cluster into the spatial core of the other, corrupting covariance estimates by pulling component means toward outlier space and destroying the optimization trajectory before β returns to 1. This directly violates Assumption 3 (Task 1.2): the method requires data whose geometric boundaries are separable under exponentiated Gaussian likelihoods. A potential fix would be replacing the Gaussian base model with a Student-t mixture, whose heavier-tailed likelihood is more robust to this corruption.

V. REFLECTION

What could not be implemented:

The BFGS and ECG baselines from the paper were not reproduced due to the complexity of the gradient computation under the repa-rameterized parameter space described in the paper's Appendix (Cholesky decomposition

$$\Sigma_j = L_j L_j^*$$

$$\alpha_j = e^{\lambda_j} / \quad {}_l e^{\lambda_l})$$

The Dirich-and soft-max reparameterization

let Process DPMM extension (Section 4) and the full MNIST clustering experiment were also excluded due to scope constraints.

What surprised me: Two things. First, the extreme sensitivity to the β-schedule: changing $\beta_{\max}$ from 1.2 to 1.5 noticeably altered convergence behavior, yet the paper provides no principled schedule-selection criterion for new datasets beyond the heuristic "more clusters → slower schedule." Second, the perturbation step is described in a single sentence in the paper, yet it required several undocumented design decisions (noise magnitude, timing, direction), and empirically without it the anti-annealing step was largely ineffective—far more impact-ful than the paper implies.

What I would revisit with more time: I would explore *adaptive* β-schedule selection driven by online estimates of mixing coefficient imbalance and cluster overlap computed from a few warm-up EM iterations—making DAAEM self-tuning in practice. I would also implement the variational Bayesian Anti-Annealing DPMM extension on high-dimensional real-world datasets (e.g., anomaly detection in biosurveillance data), where both the K-free property and the anti-annealing acceleration would be most practically valuable.

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